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Studierichting: Informatica

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$$\begin{aligned}\int_{-1}^1 \frac{1}{x^2} dx &= \int_{-1}^0 \frac{1}{x^2} dx + \int_0^1 \frac{1}{x^2} dx = \lim_{a \rightarrow 0^-} \int_{-1}^a \frac{1}{x^2} dx + \lim_{b \rightarrow 0^+} \int_b^1 \frac{1}{x^2} dx \\ &= - \lim_{a \rightarrow 0^-} \left[\frac{1}{x} \right]_{-1}^a - \lim_{b \rightarrow 0^+} \left[\frac{1}{x} \right]_b^1 = -(-\infty + 1) - (-\infty + 1) = +\infty\end{aligned}$$

References